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Class 11 BIOLOGY
NCERT BASED CHAPTER

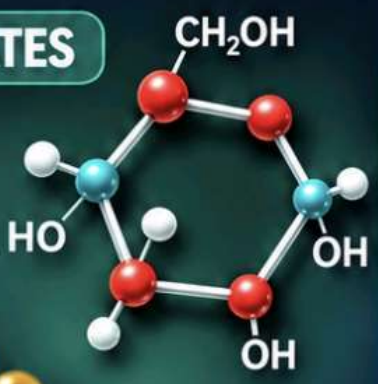
BIOMOLECULES

NEET BIOLOGY • SHORT NOTES

BIOLOGICAL MOLECULES AT A GLANCE

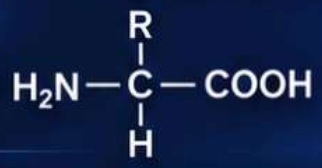
CARBOHYDRATES

- Source of energy
- Structural role
- Formed by monosaccharides



PROTEINS

- Built from amino acids
- Structural, enzymatic and regulatory roles
- Made by peptide bonds

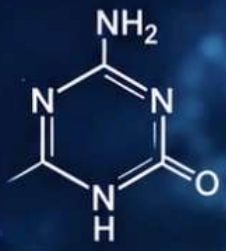


LIPIDS

- Energy rich molecules
- Water insoluble
- Includes fats, oils, phospholipids, steroids

NUCLEIC ACIDS

- Polymers of nucleotides
- Genetic material
- Types: DNA and RNA



KEY CONCEPTS | EXAM FOCUSED | EASY REVISION | NEET READY

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Class 11 Biology | NCERT Based Notes

WHAT ARE BIOMOLECULES?



Biomolecules are made up of a few chemical elements mainly **C, H, O, N, S, P** and some others like **Na, K, Ca, Mg, Fe** etc.

Biomolecules are the **naturally occurring chemical compounds** present in living organisms which are essential for their **structure, growth, metabolism and reproduction.**

TWO MAIN CATEGORIES

1. ORGANIC COMPOUNDS

- Contain carbon (C) and hydrogen (H).
- Also contain oxygen (O), nitrogen (N), sulphur (S), phosphorus (P) etc.
- Form the **major biomolecules** like **carbohydrates, proteins, lipids and nucleic acids.**

2. INORGANIC ELEMENTS / COMPOUNDS

- Do not always contain carbon.
- Present as mineral elements and simple compounds.
- **Examples:** Water, mineral ions (Na^+, K^+, Ca^{2+}, Mg^{2+}, Cl^-), inorganic compounds (NaCl, $CaCO_3$, PO_4^{3-}, SO_4^{2-}), etc.



TYPES OF BIOMOLECULES

CARBOHYDRATES

- Made of monosaccharides
- Main source of energy
- **Examples:** glucose, fructose, starch, glycogen, cellulose



PROTEINS

- Made of amino acids
- Involved in structural, enzymatic and regulatory roles
- **Examples:** enzymes, hormones, antibodies



LIPIDS

- Generally water insoluble
- Serve as energy rich molecules
- **Examples:** fats, oils, phospholipids, steroids



NUCLEIC ACIDS

- Polymers of nucleotides
- Store and transmit genetic information
- **Types:** DNA and RNA



Key Takeaway : Biomolecules are essential for life and are mainly of four types – carbohydrates, proteins, lipids and nucleic acids.

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SHORT NOTES

ELEMENTS IN LIVING ORGANISMS

All elements present in earth's crust are also present in living tissues

1 RELATIVE AMOUNT OF ELEMENTS

Element	% Weight of	
	Earth's crust	Human body
Hydrogen (H)	0.14	0.5
Carbon (C)	0.03	18.5
Oxygen (O)	46.6	65.0
Nitrogen (N)	very little	3.3
Sulphur (S)	0.03	0.3
Sodium (Na)	2.8	0.2
Calcium (Ca)	3.6	1.5
Magnesium (Mg)	2.1	0.1
Silicon (Si)	27.7	negligible

The relative abundance of **carbon** and **hydrogen** with respect to other elements is higher in any living organism than in earth's crust.

2 TYPES OF CONSTITUENTS

Living organisms / Tissues

Carbon / Organic compounds (Biomolecules)

Chemistry point of view:
- Functional groups like aldehyde, ketones, aromatic compounds, etc.

Inorganic elements / compounds

Biological point of view:
- Amino acids, nucleotide bases, fatty acids, etc.

3 IMPORTANT INORGANIC COMPONENTS

Component	Formula / Ions
Sodium	Na^+
Potassium	K^+
Calcium	Ca^{2+}
Magnesium	Mg^{2+}
Water	H_2O
Compounds	NaCl, $CaCO_3$ PO_4^{3-}, SO_4^{2-}

Component	% of total cellular mass
Water	70 – 90
Proteins	10 – 15
Carbohydrates	3
Lipids	2
Nucleic acids	5 – 7
Ions	1

4 WHERE DO WE FIND THESE ELEMENTS?

Present in both – Earth's crust and living tissues

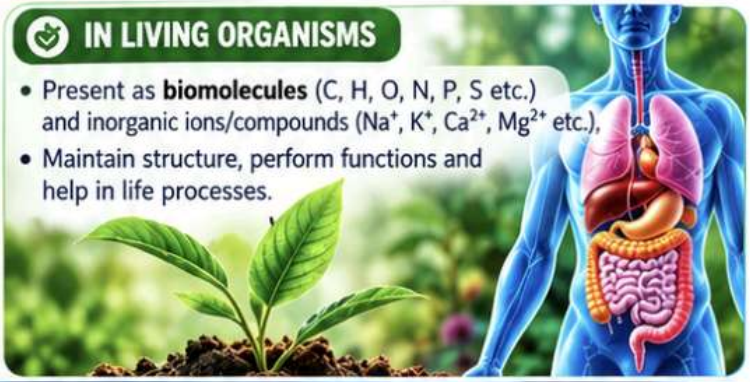
IN EARTH'S CRUST

- Major elements: O, Si, Al, Fe, Ca, Na, K, Mg, etc.
- Rock minerals, soil and water contain these elements in various forms.



IN LIVING ORGANISMS

- Present as **biomolecules** (C, H, O, N, P, S etc.) and inorganic ions/compounds (Na^+, K^+, Ca^{2+}, Mg^{2+} etc.).
- Maintain structure, perform functions and help in life processes.



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CHEMICAL COMPOSITION OF LIVING TISSUES

Two ways to decode what a living system is made of

1 ELEMENTAL ANALYSIS

It gives the elemental composition of living tissues (H, C, O, N, S, Na, Ca, Mg, etc.)

1 Living tissue (a leaf or liver)



Wet weight (as it is)

2 Evaporation of water (drying)



Dry weight (after removing all water)

3 Complete burning (oxidation)



Ash

Inorganic elements (like Ca, Mg, Na, K, etc.)

Complete burning oxidises carbon compounds to gaseous products such as CO_2 and water vapour.

2 CHEMICAL ANALYSIS

It gives an idea of the kind of organic and inorganic constituents present in living tissues.

Living tissue (a piece of liver/ vegetable, etc.)



Grinding



Thick slurry

Add Trichloroacetic acid (TCA)



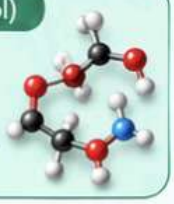
(Cl_3CCOOH)

Filtration



FILTRATE (Acid-soluble pool)

- Mostly small biomolecules + inorganic compounds – 18 – 800 Da
- Examples: sugars, amino acids, nucleotides, salts, etc.



RETENTATE (Acid-insoluble fraction)

- Macromolecules + organelles – 10,000 Da and above (lipids are an exception)
- Examples: proteins, nucleic acids, polysaccharides, organelles, etc.



NEET HIGH-YIELD

1. **Chemical analysis** identifies organic & inorganic constituents present in living tissues.
2. **Elemental analysis** reveals the elemental composition of living tissues (H, C, O, N, S, etc.).
3. **Lipids** are generally not polymers and are not strictly macromolecules.

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BIOMOLECULES

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NCERT BASED SHORT NOTES

METABOLITES

Biomolecules found in living systems

1 PRIMARY METABOLITES

- Have identifiable functions.
- Play known roles in normal physiological processes.

Essential for life processes

Examples:

- Sugars (carbohydrates)
- Amino acids
- Fats & oils (lipids)
- Nitrogen bases
- Nucleosides
- Nucleotides



2 SECONDARY METABOLITES

- Roles/functions in host organisms may not be fully understood.
- Many are useful to human welfare.
- Some have ecological importance.

Important for human welfare & ecological balance

Examples:

- Alkaloids
- Flavonoids
- Rubber
- Essential oils
- Antibiotics
- Coloured pigments
- Scents
- Gums
- Spices



HIGH-YIELD EXAMPLES OF SECONDARY METABOLITES

Pigments
- Carotenoids
- Anthocyanins



Alkaloids
- Morphine
- Codeine



Terpenoids
- Monoterpenes
- Diterpenes



Essential oils
- Lemongrass oil



Toxins
- Abrin
- Ricin



Lectins
- Concanavalin A



Drugs
- Vinblastine
- Curcumin



Polymeric substances
- Rubber
- Gums
- Cellulose



POLYSACCHARIDES

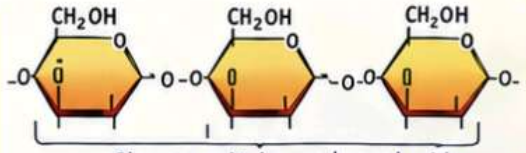
Long chains of sugars

Homopolysaccharides

- Consist of only one type of monomer unit repeating 'n' times.
- **Example:** Cellulose, etc.

Heteropolysaccharides

- Consist of different monomer units.
- **Examples:** Starch, Glycogen, Inulin, Chitin, etc.



Glucose units in a polysaccharide (e.g. Starch, Cellulose)

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BIOMOLECULES

POLYSACCHARIDES

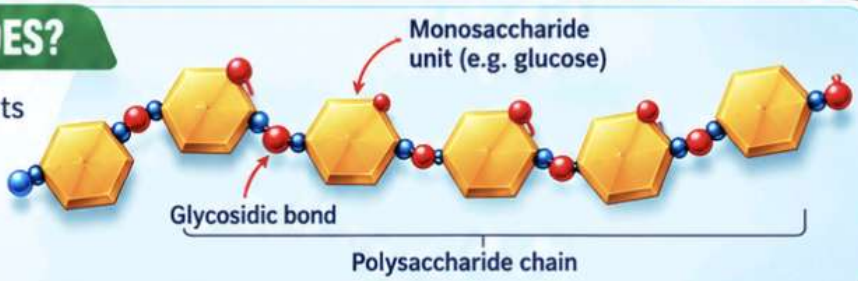
LONG CHAINS OF SUGAR UNITS

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NCERT BASED SHORT NOTES

WHAT ARE POLYSACCHARIDES?

- Long chains of sugar (monosaccharide) units linked together by glycosidic bonds.
- They are generally high molecular weight biomolecules.
- Mostly water insoluble.



TYPES OF POLYSACCHARIDES

Homopolysaccharides

- Consist of only one type of monomer unit repeating 'n' number of times.
- **Examples:** Cellulose, Starch, Glycogen, Inulin.



Same monomer unit

Heteropolysaccharides

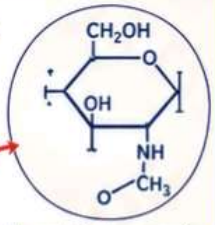
- Consist of different monomer units.
- **Examples:** Chitin, Hyaluronic acid, etc.



Different monomer units

Complex polysacharides

- May contain amino-sugars and chemically modified sugars.
- Mostly heteropolymers.
- **Example:** Chitin (contains N-acetyl glucosamine).



N-acetyl glucosamine (unit of chitin)



IMPORTANT POLYSACCHARIDES – COMPARISON TABLE

Polysaccharide	Found in	Monomer units	Function	
Cellulose	Plants	Glucose	Formation of cell wall (structural role)	
Starch	Plants	Glucose	Storage of energy	
Glycogen	Animals	Glucose	Storage of energy	
Inulin	Plants	Fructose	–	
Chitin	Animals (Arthropods)	N-acetyl glucosamine	Formation of exoskeleton in arthropods	

NEET TRAPS & QUICK FACTS

1. **Cellulose** → structural polysaccharide (plant cell wall).
2. **Starch & Glycogen** → storage polysaccharides (plants and animals respectively).
3. **Inulin** → polymer of fructose (in plants).
4. **Chitin** → contains N-acetyl glucosamine (found in arthropods).
5. Polysaccharides are generally water insoluble.

Important for NEET!

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BIOMOLECULES

LIPIDS

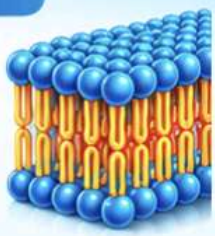
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NCERT BASED SHORT NOTES

Generally water insoluble organic compounds

1 WHAT ARE LIPIDS?

- Generally water insoluble.
- Not polymeric substances.
- Molecular weight: Not more than 800 D.
- Not strictly biomacromolecules.
- Arranged to form cell membrane and other membranes.
- Pieces form vesicles (water insoluble).



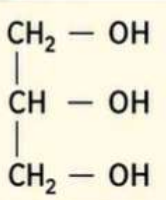
Lipid bilayer in cell membrane

2 TYPES OF LIPIDS

Lipids

Simple fatty acids	Glycerides (Glycerol + fatty acids)	Phospholipids (Glycerol + fatty acids + phosphorylated group)
 R—COOH	• Monoglycerides • Diglycerides • Triglycerides	**Example:** Lecithin

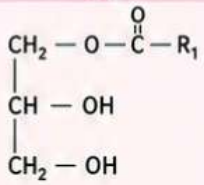
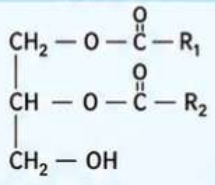
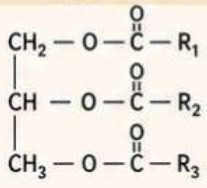
3 GLYCEROL



Glycerol

- A three carbon compound with three –OH groups.
- Combines with fatty acids to form glycerides.

4 GLYCERIDES (FATTY ACIDS ESTERIFIED WITH GLYCEROL)

Monoglyceride	Diglyceride	Triglyceride
		

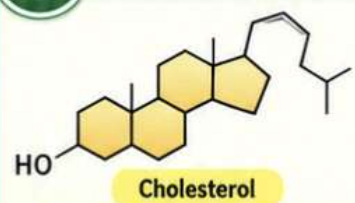
R_1, R_2 and R_3 are fatty acids.

5 PHOSPHOLIPIDS



Phospholipid (Lecithin)

6 CHOLESTEROL



Cholesterol

- A sterol found in animal tissues.
- Some tissues especially neural tissues have lipids with more complex structures.

7 FATTY ACIDS

- General formula: R – COOH
- R group can be:

$CH_3-(CH_2)_{n-2}-COOH$ Palmitic acid

Methyl (CH_3)	Ethyl (C_2H_5)	Higher number of $-CH_2$ groups (↑ to 19 C)

- **Examples:**
- Palmitic acid (16 C including carboxyl carbon).
- Arachidonic acid (20 C including carboxyl carbon).

8 TYPES OF FATTY ACIDS

Feature	Saturated fatty acids	Unsaturated fatty acids
C = C double bonds	Absent (×)	One or more (✓)
Structure	Straight chain	Kinked chain
Examples	Palmitic acid, Stearic acid	Oleic acid, Linoleic acid, Arachidonic acid

NEET QUICK POINTS

1. Lipids are generally water insoluble.
2. Not polymeric and molecular weight < 800 D.
3. Glycerol + fatty acids → glycerides (mono, di, tri).
4. Phospholipids (e.g. lecithin) are major components of cell membranes.
5. Cholesterol is a sterol present in animal tissues.

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BIOMOLECULES

NUCLEIC ACIDS

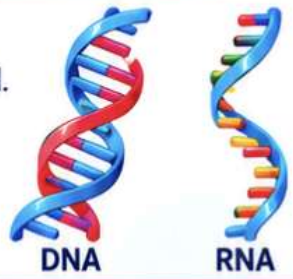
Genetic material of living organisms

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NCERT BASED SHORT NOTES

1 WHAT ARE NUCLEIC ACIDS?

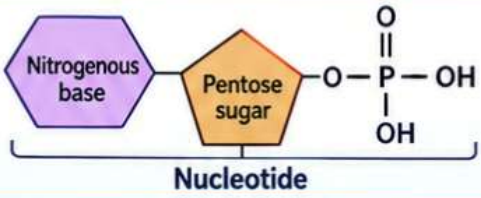
- **Examples:** DNA and RNA.
- Function as genetic material.
- They are polynucleotides (polymers of nucleotides).



2 NUCLEOTIDE – BUILDING BLOCK

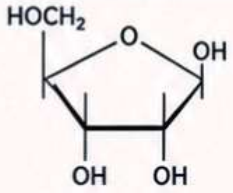
Nucleotide has 3 components

- **Nitrogenous base** (Heterocyclic compound)
- **Pentose sugar** (Ribose (in RNA) 2' Deoxyribose (in DNA)
- **Phosphate group** (Phosphoric acid)



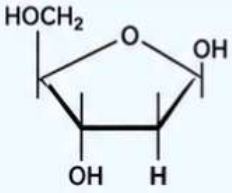
3 PENTOSE SUGARS

Ribose (in RNA)



$C_5H_{10}O_5$
(Ribose)

2'-Deoxyribose (in DNA)

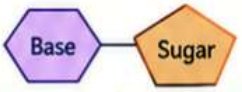


$C_5H_{10}O_4$
(2'-Deoxyribose)

4 NUCLEOSIDE vs NUCLEOTIDE

Nucleoside

- Nitrogenous base + sugar
- No phosphate group



Nucleoside

Nucleotide

- Nitrogenous base + sugar + phosphate group



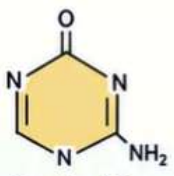
Nucleotide

5 NITROGENOUS BASES

Purines (Double ring)



Adenine (A)

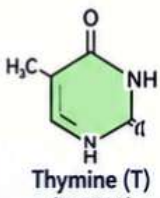


Guanine (G)

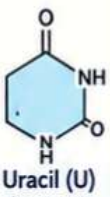
Pyrimidines (Single ring)



Cytosine (C)



Thymine (T) (in DNA)



Uracil (U) (in RNA)

Nucleoside and Nucleotide names

Nitrogenous base	Nucleoside	Nucleotide
Adenine	Adenosine	Adenylic acid
Guanine	Guanosine	Guanylic acid
Cytosine	Cytidine	Cytidylic acid
Thymine	Thymidine	Thymidylic acid
Uracil	Uridine	Uridylic acid

NEET KEY POINTS

1. Nucleic acids are polynucleotides.
2. Nucleotide = Base + Sugar + Phosphate.
3. Sugar: Ribose (RNA) or 2'-deoxyribose (DNA).
4. Bases: Purines (A, G) and Pyrimidines (C, T, U).
5. RNA has Uracil, DNA has Thymine.

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BIOMOLECULES

DNA vs RNA

Nucleic Acids • NEET Quick Revision

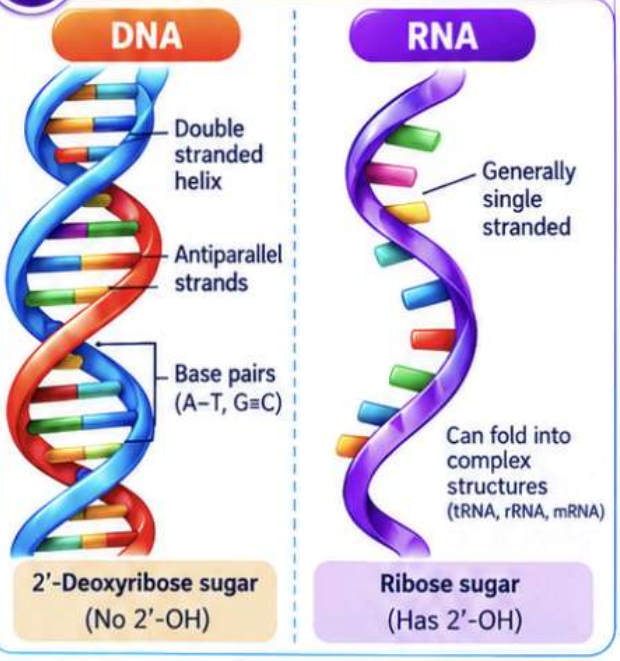
Class 11 BIOLOGY

NCERT BASED SHORT NOTES

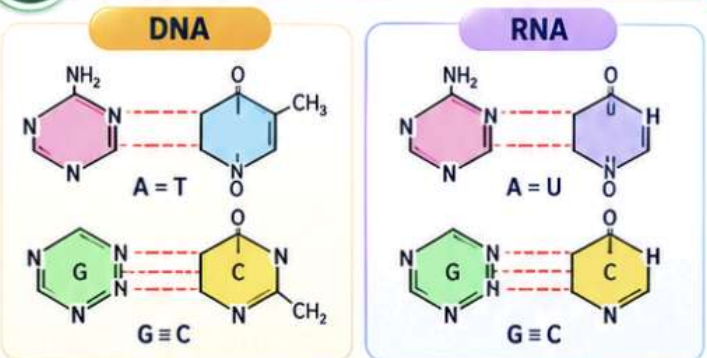
1 DNA vs RNA : KEY DIFFERENCES

Feature	DNA	RNA
Sugar	2′-Deoxyribose ($C_5H_{10}O_4$)	Ribose ($C_5H_{10}O_5$)
Nitrogenous bases	Adenine (A), Guanine (G), Cytosine (C), Thymine (T)	Adenine (A), Guanine (G), Cytosine (C), Uracil (U)
Typical structure	Double-stranded (antiparallel helix)	Generally single-stranded (can fold into complex structures)
Main role	Genetic material, information storage	Genetic information expression and protein synthesis
Stability	More stable (lacks 2′-OH group)	Less stable (2′-OH group makes it more reactive)
Location / general occurrence	Mainly in nucleus, also in mitochondria and chloroplasts	In nucleus, cytoplasm and ribosomes

2 STRUCTURE



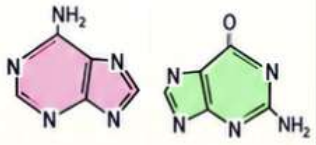
3 BASE PAIRING



In both DNA and RNA, G pairs with C (3 hydrogen bonds) which is stronger than A–T (2 hydrogen bonds) or A–U (2 hydrogen bonds).

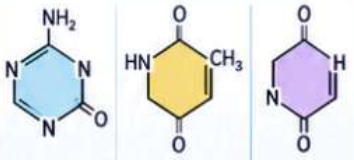
4 PURINES vs PYRIMIDINES

Purines (Double ring structure)



Adenine (A) | Guanine (G)

Pyrimidines (Single ring structure)



Cytosine (C) | Thymine (T) (in DNA) | Uracil (U) (in RNA)

5 KEY POINTS

1. Nucleic acids are polynucleotides.
2. **Nucleotide** = Base + Sugar + Phosphate.
3. Sugar: Ribose (RNA) or 2′-deoxyribose (DNA).
4. Bases: Purines (A, G) and Pyrimidines (C, T, U).
5. RNA has Uracil, DNA has Thymine.

7 NEET TRAPS

- **Nucleotide** = Nitrogenous base + Sugar + Phosphate group
- **Nucleoside** = Nitrogenous base + Sugar (without phosphate group)
- Both DNA and RNA are polynucleotides.
- **Purines:** A, G | **Pyrimidines:** C, T (DNA), U (RNA).

Important for NEET!

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BIOMOLECULES

PROTEINS & AMINO ACIDS

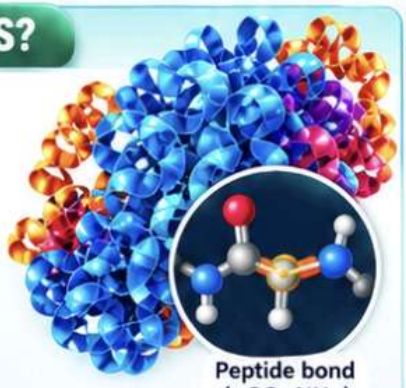
Class 11 BIOLOGY

NCERT BASED SHORT NOTES

CLASS 11 BIOLOGY • NEET SHORT NOTES

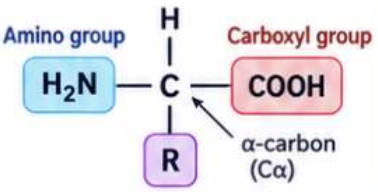
1 WHAT ARE PROTEINS?

- Proteins are **polypeptides**, i.e. long chains of amino acids linked by **peptide bonds**.
- They are **heteropolymers** of amino acids.
- Proteins perform a wide variety of structural, catalytic, transport, regulatory and protective functions in living organisms.



Peptide bond (–CO–NH–)

2 GENERAL STRUCTURE OF AMINO ACID



Each amino acid has an α-carbon attached to $—NH_2$, —COOH, —H and a variable side chain (R group).

Examples of R group (Side chain)

Glycine (Gly)	Alanine (Ala)	Serine (Ser)
R = H	$R = CH_3$	$R = CH_2OH$
$H_2N—CH(H)—COOH$	$H_2N—CH(CH_3)—COOH$	$H_2N—CH(CH_2OH)—COOH$

3 IONISATION AND ZWITTERIONIC FORM

- Amino acids have ionisable amino ($–NH_2$) and carboxyl (–COOH) groups.
- In aqueous solution, they exist mainly as **zwitterions** (dipolar ions).

$H_3\overset{+}{N}—CH(R)—COO^-$

Zwitterionic form (at neutral pH)

4 CLASSIFICATION BASED ON NATURE OF R GROUP

Acidic amino acids (negatively charged R group)	Basic amino acids (positively charged R group)	Neutral amino acids (non-polar R group)
Example: Glutamic acid (Glu)	**Example:** Lysine (Lys)	**Example:** Valine (Val)
$H_2N—CH(CH_2—CH_2—COOH)—COOH$	$H_2N—CH(R=(CH_2)_4—NH_2)—COOH$	$H_2N—CH(CH(CH_3)_2)—COOH$

5 CLASSIFICATION BY BODY REQUIREMENT

Essential amino acids

- Cannot be made by the body.
- Must be obtained from diet.



Non-essential amino acids

- Can be made by the body from simple precursors.



6 AROMATIC AMINO ACIDS

Contain aromatic ring in their R group.

- **Examples:** Tyrosine, Phenylalanine, Tryptophan.

Tyrosine (Tyr)	Phenylalanine (Phe)	Tryptophan (Trp)
CH_2—(benzene)—OH	CH_2—(benzene)	(indole)—CH_2

NEET HIGH-YIELD POINTS

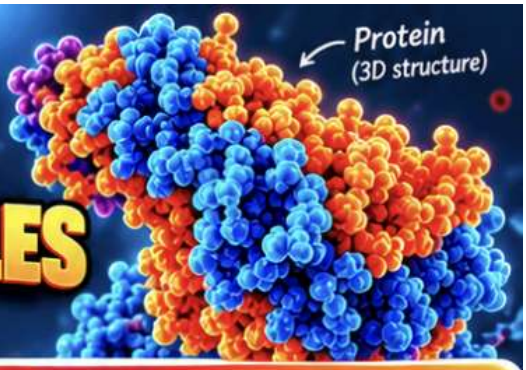
1. Proteins are **polypeptides** (heteropolymers of amino acids) linked by peptide bonds.
2. **Each amino** acid has an α-carbon attached to $—NH_2$, —COOH, —H and R group.
3. Amino acids are zwitterions in solution: $H_3N^+—CH(R)—COO^-$.
4. 20 amino acids occur in proteins.
5. **Acidic:** Glutamic acid | **Basic:** Lysine | **Neutral:** Valine.
6. **Essential:** cannot be made by body, required in diet.
7. **Non-essential:** can be made by body.
8. **Aromatic amino acids:** Tyrosine, Phenylalanine, Tryptophan.

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BIOMOLECULES

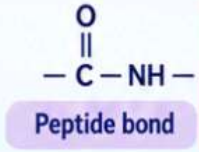
PROTEINS & FUNCTIONS & EXAMPLES

Class 11 Biology • NEET Short Notes



WHAT ARE PROTEINS?

- Proteins are **polypeptides**, i.e. long chains of **amino acids.**
- They are **heteropolymers** of amino acids linked by **peptide bonds** (–CO–NH–).



KEY POINTS

- Made up of 20 different amino acids.
- Perform a wide variety of structural, catalytic, transport, regulatory and protective functions.
- Present in all living cells.

Each amino acid has an α-carbon attached to $-NH_2$, –COOH, –H and a variable side chain (R group).

MAJOR FUNCTIONS OF PROTEINS

1 Structural Function



Provide strength, shape and support to cells and tissues.

2 Enzymatic Function



Act as biological catalysts and speed up reactions.

3 Hormonal / Regulatory Function



Act as hormones to regulate body processes.

4 Transport Function



Transport molecules in body fluids and across membranes.

5 Defense Function



Help in fighting infectious agents (antibodies).

6 Receptor Function



Receive and transduce signals (sensory reception).

7 Membrane Transport



Channel or carrier proteins help in transport across membranes.

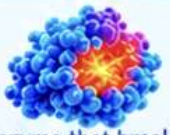
IMPORTANT EXAMPLES

Collagen



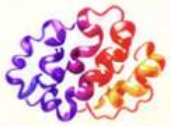
Major structural protein of intercellular ground substance (connective tissue).

Trypsin



Enzyme that breaks down proteins into smaller peptides during digestion.

Insulin



Hormone that regulates blood glucose level.

Antibody (Immunoglobulin)



Helps in fighting infectious agents (pathogens).

Receptor protein



Involved in sensory reception (smell, taste, hormones, etc.).

GLUT-4

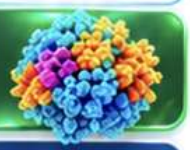


Enables glucose transport into cells (especially in muscle and adipose tissue).

1 **Collagen** is the most abundant protein in the animal world.



2 **RuBisCO** is the most abundant protein in the whole biosphere (especially in plants).



NEET QUICK RECALL

1. Proteins are **polypeptides** (heteropolymers of amino acids) linked by **peptide bonds.**
2. Each amino acid has an α-carbon attached to $-NH_2$, –COOH, –H and R group.
3. Amino acids are zwitterions in solution: $H_3N^+–CH(R)–COO^-$.
4. 20 amino acids occur in proteins.
5. **Acidic:** Glutamic acid | **Basic:** Lysine | **Neutral:** Valine.
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BIOMOLECULES

LEVELS OF PROTEIN STRUCTURE

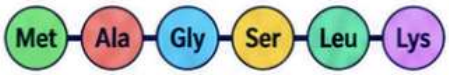
From Amino Acid Sequence to Functional Protein

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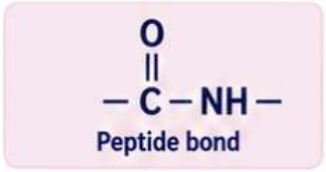
NCERT BASED SHORT NOTES

1 PRIMARY STRUCTURE

- Linear sequence of amino acids in a polypeptide chain.
- Amino acids are linked by peptide bonds (–CO–NH–).
- Determines the higher levels of structure.



Amino acid sequence (primary structure)

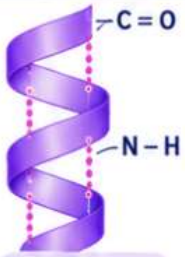


Peptide bond

Example: Insulin (A chain) ⟶ Gly – Ile – Val – Glu – Gln – Cys ...

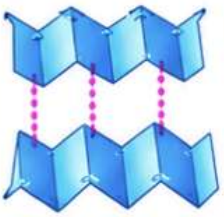
2 SECONDARY STRUCTURE

- Local folding of polypeptide chain.
- Stabilized by hydrogen bonds between C = O and N – H groups of peptide bonds.
- Two main types: α-Helix and β-Pleated sheet.



α-Helix

Right-handed coil stabilized by intra-chain hydrogen bonds.



β-Pleated Sheet

Extended zig-zag structure stabilized by inter-chain hydrogen bonds.

3 TERTIARY STRUCTURE

- Overall 3D folding of a single polypeptide chain.
- Stabilized by various interactions:
  - Hydrogen bonds
  - Ionic bonds (salt bridges)
  - Disulfide bonds (–S–S–)
  - Hydrophobic interactions
  - Van der Waals forces

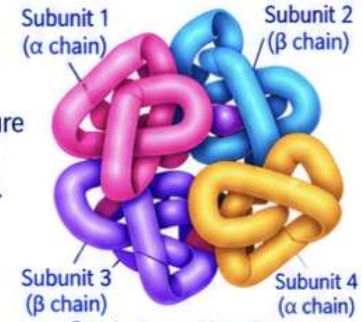


Tertiary structure (of one polypeptide chain)

Example: Myoglobin (globular protein) → compact 3D structure.

4 QUATERNARY STRUCTURE

- Association of two or more polypeptide chains (subunits).
- Stabilized by same types of interactions as tertiary structure (hydrogen bonds, ionic bonds, hydrophobic interactions, etc.).
- Each subunit is called a protomer.
- Not present in all proteins.



Quaternary structure (e.g., Hemoglobin)

Example: Hemoglobin – 4 subunits (2 α + 2 β).

5 COMPARISON TABLE

Level	What it shows?	Stabilizing forces	Example
Primary	Sequence of amino acids	Peptide bonds	Insulin (A chain)
Secondary	Local folding (α-helix, β-sheet)	Hydrogen bonds	Keratin (α-helix)
Tertiary	Overall 3D structure (single chain)	Hydrogen, ionic, hydrophobic, disulfide bonds	Myoglobin
Quaternary	Association of multiple chains	Same as tertiary	Hemoglobin (2α + 2β)

6 NEET KEY POINTS

1. Primary structure = sequence of amino acids.
2. Secondary structure = α-helix and β-pleated sheet.
3. Tertiary structure = 3D folding of single chain.
4. Quaternary structure = association of multiple subunits.
5. Stabilizing forces: H-bonds, ionic bonds, hydrophobic interactions, disulfide bonds, van der Waals forces.
6. Hemoglobin is an example of quaternary structure.

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